

The recommended word count for this criterion is **300 words**.

Marks	Level descriptor
0	The response does not reach a standard described by the descriptors below.
1–2	The response: • outlines a problem scenario • states limited success criteria • outlines the nature of the solution in a computational context.
3–4	The response: • describes the problem scenario in terms of its measurable solution requirements • states appropriate success criteria • explains the choice of computational context for the solution.

Problem specification clarifications

The **problem scenario** is a clear description of the problem including its measurable solution requirements. The description should relate directly to the problem, whether this be in the world around us, other fields of knowledge, or a current issue in computing.

Success criteria are measurable outcomes derived from the solution requirements that indicate the successful development of the product.

The **computational context** is the specific area of computing that is selected to be used in the solution.

Criterion B: Planning (4 marks)

The planning of the product must be consistent with the problem specification in criterion A.

- This criterion assesses how the problem scenario has been decomposed into component parts.
- The plan should address the requirements of the solution in terms of the success criteria, and include a proposed chronology for the steps involved in planning, designing, developing, testing and evaluating the solution.
- A plan can be presented in different forms, but popular diagram formats such as GANTT and AGILE charts can effectively support the planning process.
- The plan may include any relevant research, such as the use of existing code libraries.

The recommended word count for this criterion is **150 words**.

Marks	Level descriptor
0	The response does not reach a standard described by the descriptors below.
1–2	The response: • constructs a partial decomposition of the problem scenario • constructs a plan that addresses some of the success criteria of the solution.
3–4	The response: • constructs a reasonable decomposition of the problem scenario • constructs a plan that addresses the success criteria of the solution.

Planning clarifications

Decomposition is the breaking down of the problem scenario identified in criterion A into smaller, more manageable sub-problems or components. The decomposition can be effectively constructed using diagrams.

A **reasonable decomposition** breaks the problem down into essential components that support the construction of a plan.

Criterion C: System overview (6 marks)

The system overview of the product must be consistent with the problem specification in criterion A and the planning in criterion B.

- The system overview should include a system model with the key components, their relationships, the rules governing their interaction, and the algorithms required by these components and the user interface.
- The system overview should have the clarity to enable a third party to recreate the product.
- The system model will provide the information for a viable testing strategy.

The recommended word count for this criterion is **150 words**.

Marks	Level descriptor
0	The response does not reach a standard described by the descriptors below.
1–2	The response: • outlines a limited system model • identifies algorithms for the components of the system model • identifies a testing strategy for at least one success criterion.
3–4	The response: • constructs a system model that is not complete • constructs algorithms for the components of the system model that lead to partial functionality of the product • outlines a testing strategy that aligns with at least three success criteria.
5–6	The response: • constructs a complete system model • constructs algorithms for the components of the system model that enable the product to perform • describes a testing strategy that aligns with the success criteria.

System overview clarifications

A **system model** consists of diagrams that include the components of the system and how they are connected. The system model will include the design of the user interface. A **complete system model** does not include the algorithms for each of the components.

Algorithms can be presented in different forms, including natural language, flow charts or pseudocode, and should address the individual components of the system model.

The **testing strategy** refers to a systematic approach for evaluating whether the computational solution works as intended. The testing strategy should ensure that code functions correctly and handles

System overview clarifications

unexpected or incorrect inputs. This can be represented effectively in a table with proposed test data and expected outcomes.

Criterion D: Development (12 marks)

The development of the product must be consistent with the problem specification in criterion A, the planning in criterion B and the system overview developed in criterion C.

- The video must provide evidence of the functionality and give examples of the testing of the product.
- The development of the solution must justify the structure of the product, why the structure is appropriate, and demonstrate the techniques used to develop the product based on the algorithms constructed in criterion C. These techniques may include loops, data structures, existing libraries and the integration of software tools.
- The testing strategy must include testing for correctness, reliability and efficiency. The testing must be described and justified in the documentation with supporting examples seen in the video.

The recommended word count for this criterion is **1,000 words**.

Marks	Level descriptor
0	The response does not reach a standard described by the descriptors below.
1–3	The response: • constructs a product with very limited functionality • constructs a product using no appropriate techniques to implement the algorithms • states the choices made to implement the algorithms • states the testing strategy used.
4–6	The response: • constructs a product that has limited functionality • constructs a product using at least one appropriate technique to implement the algorithms • outlines the choices made to implement the algorithms • states the effectiveness of the testing strategy.
7–9	The response: • constructs a product that has partial functionality • constructs a product that uses some appropriate techniques to implement the algorithms • explains the choices made to implement the algorithms • describes the effectiveness of the testing strategy.
10–12	The response: • constructs a fully functional product • constructs a product that uses appropriate techniques to implement the algorithms • evaluates the choices made to implement the algorithms • justifies the effectiveness of the testing strategy.

Development clarifications

Implementation and coding of the algorithms: Techniques in the criteria refer to the process of programming algorithms using code. The documentation must highlight key elements of code that are important for the efficient functioning of the algorithms. Any code presented in the solution must include relevant comments, be consistent and be readable. Code excerpts included in the documentation must be referenced to the full source code submitted as an appendix.

The video must demonstrate the **functionality** of the product. The deployment of the **testing** strategy and its effectiveness must be described in the documentation, with examples of the testing seen in the video.

Criterion E: Evaluation (4 marks)

The evaluation of the product must be consistent with the problem specification and success criteria in criterion A.

The recommended word count for this criterion is **400 words**.

Marks	Level descriptor
0	The response does not reach a standard described by the descriptors below.
1–2	The response: • states the extent to which the success criteria were met • describes improvements to the product.
3–4	The response: • evaluates the extent to which the success criteria were met • justifies improvements to the product.